

GTH TechVerse 2026

Future Skills Bootcamp

Learn • Build • Innovate

Organized By

Geeta Technical Hub (GTH) Team

In Association With

- School of Computer Science & Engineering

Vision

The transition from school to university is one of the most important phases in a student's journey. To make this transition exciting, engaging, and industry-oriented, Geeta Technical Hub (GTH) proposes a **3-Day Practical Future Skills Bootcamp** for newly admitted students before the official Orientation Program.

Unlike traditional orientation sessions, this bootcamp will provide students with real-world exposure to emerging technologies through completely practical learning experiences while simultaneously developing communication skills, leadership, teamwork, creativity, and digital presence.

The objective is to create an innovation-driven culture from the very first day of a student's university life.

Objectives

- Introduce students to emerging technologies through practical learning.
- Develop confidence before the academic session begins.
- Build a strong student community from Day One.
- Enhance communication and leadership skills.
- Promote teamwork and collaboration.
- Encourage innovation and problem-solving.
- Increase university branding through student-generated digital content.
- Support admission outreach through structured referral activities.
- Create excitement around Geeta University and innovation culture.

Bootcamp Duration

Duration: 3 Days

Timing: 10:00 AM – 4:00 PM

Mode: 100% Practical Learning

Target Audience: Newly Admitted Students & Prospective Students

Bootcamp Structure

The bootcamp consists of four integrated learning tracks:

- Internet of Things (IoT)
- Artificial Intelligence (AI)
- Cyber Security
- Communication, Leadership & Community Building

Every day combines technical learning with practical activities, teamwork, competitions and community engagement.

Bootcamp Schedule

Day 1 – IoT Foundation & Community Building

Welcome Session

- Welcome & Registration
- Introduction to Geeta Technical Hub (GTH)
- Bootcamp Overview
- Team Formation

Ice Breaking & Community Building

Students will participate in fun and engaging activities designed to help them connect with their peers and build confidence.

Activities

- Ice Breaking Games
 - Team Building Activities
 - Communication Challenges
 - Collaboration Tasks
 - Leadership Games
 - Problem Solving Activities
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Internet of Things (IoT)

Students will begin their IoT journey through completely practical sessions using ESP32 and Artificial Intelligence.

Project 1 – Gesture Control Servo

Students will learn how Artificial Intelligence can recognize hand gestures using a smartphone camera and control a servo motor through ESP32 in real time.

Learning Outcomes

- ESP32 Programming
 - Computer Vision
 - Python Hand Tracking
 - Wireless Communication
 - Servo Control
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Project 2 – AI Face Controlled RGB Light

Students will build an AI-powered emotion recognition system.

Features

- Smile → Green LED
- Angry → Red LED
- Surprised → Blue LED

Daily Community Challenge

Students will earn leaderboard points through:

- Team Collaboration
 - Project Completion
 - Helping Other Teams
 - Communication Activities
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Day 2 – Artificial Intelligence, IoT & Community Challenges

Artificial Intelligence (AI)

Students will explore modern AI tools through completely practical sessions.

Topics

- Prompt Engineering
- AI Coding
- AI Website Development
- AI Presentation Creation
- AI Image Generation
- AI Video Generation
- AI Research Tools
- AI Productivity
- AI Automation

Students will complete hands-on AI challenges throughout the day.

Internet of Things (IoT)

Flagship Project – GenVision AI

Students will build an AI Vision platform capable of controlling IoT devices using only a smartphone camera.

Modules

- Hand Gesture Recognition
- Face Detection
- Face Tracking
- Pose Detection
- Object Detection
- ESP32 Communication
- Mobile App Control

Students will experience how Artificial Intelligence can directly control physical hardware.

Community Building Activities

Students will participate in:

- Innovation Challenges
 - Team Problem Solving
 - Leadership Activities
 - Communication Games
 - Networking Sessions
 - Social Media Challenges
 - Daily Leaderboard Competition
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Day 3 – Advanced IoT, Cyber Security & Grand Finale

Final IoT Project – NeuroVision

Students will build an advanced Human-Machine Interaction project using an **EMG (Electromyography) Muscle Sensor**.

Instead of using a keyboard or mouse, students will control a custom-built **Dino Escape** game using their own muscle signals.

Project Features

- EMG Muscle Signal Detection
- Real-Time Signal Processing
- Muscle Controlled Dino Escape Game
- Jump using Muscle Flex
- Long Jump using Strong Muscle Flex
- Slide using Double Muscle Flex
- Live Muscle Activity Monitoring
- Human-Machine Interaction

Students will gain practical exposure to embedded systems, biomedical sensing, signal processing, and interactive game development.

Cyber Security Practical Session

Students will learn cyber security through practical demonstrations and awareness activities.

Topics Covered

- Password Security
 - Brute Force Demonstration
 - Phishing Awareness
 - Digital Footprint Analysis
 - QR Code Security
 - Wi-Fi Security Basics
 - Safe Browsing Practices
 - USB Device Awareness
 - Data Privacy & Protection
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Final Project Showcase

Each team will present their projects before mentors and judges.

Projects

- Gesture Control Servo
 - AI Face Controlled RGB Light
 - GenVision AI
 - NeuroVision (EMG Controlled Dino Escape)
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Grand Finale

Technical Challenge

Students will participate in the final innovation challenge.

Innovation Quiz

A fun technical quiz based on concepts learned during the bootcamp.

Leaderboard Announcement

Final scores will be announced based on:

- Technical Skills
 - Project Completion
 - Innovation
 - Community Participation
 - Communication Activities
 - Daily Challenges
 - Referral Points
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Prize Distribution

Awards include:

Overall Champion

Runner-Up

Best Community Leader

Certificates of Completion for All Participants

Closing Ceremony

- Bootcamp Highlights
- Mentor Feedback
- Student Experience Sharing
- Trophy Distribution
- Certificate Distribution
- Group Photograph
- Closing Address

Gamified Learning System

A live leaderboard will be maintained throughout all three days.

Students earn points through:

- Project Completion
- Daily Challenges
- Innovation Tasks
- Leadership Activities
- Communication Challenges
- Project Presentation

Community Engagement Challenges

Students can earn additional points by completing branding activities such as:

- Posting on LinkedIn
- Sharing Instagram Stories
- Creating Instagram Reels
- Uploading project demonstrations
- Sharing learning experiences
- Tagging Geeta University
- Promoting innovation culture

Referral Challenge

Students receive bonus points for successful referrals.

Tasks include:

- Sharing the GUTS Examination registration link.
- Encouraging prospective students to register.
- Bonus points for every successful registration with fee payment.

Rewards & Recognition

Top-performing students will receive:

-  Overall Champion Trophy
-  Runner-Up Trophy
-  Best Community Leader
-  Certificates of Completion
-  Digital Achievement Badges

Every participant will receive an official participation certificate.

Required Infrastructure

- Computer Lab
- High-Speed Internet
- ESP32 Development Boards
- Servo Motors
- RGB LEDs
- Breadboards & Sensors

Expected Outcomes

By the end of the bootcamp, students will:

- Build real-world IoT projects.
- Experience AI-powered hardware development.
- Learn practical cyber security fundamentals.
- Improve communication and presentation skills.
- Develop teamwork and leadership abilities.
- Become active members of the university community.
- Gain confidence before beginning academic classes.
- Understand the innovation ecosystem at Geeta University.
- Promote the university through positive digital engagement.

Benefits to Geeta University

- Creates a strong first impression before orientation.
- Builds excitement among newly admitted students.
- Promotes Geeta Technical Hub (GTH) and innovation culture.
- Increases university branding through student-generated content.
- Encourages collaboration and networking.
- Supports admission outreach through referral activities.
- Introduces students to industry-relevant technologies from Day One.
- Establishes an innovation-driven campus environment.

Conclusion

GTH TechVerse 2026 – Future Skills Bootcamp is a **3-day immersive learning experience** designed to transform newly admitted students into confident, collaborative, and innovation-driven learners before the start of their academic journey.

Through practical learning in **IoT, Artificial Intelligence, Cyber Security, Communication Skills, Leadership Development, Community Building, and Gamified Challenges**, students will not only gain technical knowledge but also build meaningful connections, develop professional skills, and become active ambassadors of Geeta University.

The program aims to create a memorable first experience, laying the foundation for a strong innovation ecosystem and a vibrant student community from the very beginning.